

Scratch: Loops — Homework Worksheet

Due Mon, Jul 6 · Session 4

Build a looping pattern or animation · Log your work · Think like an AI!

🕒 30 MIN TOTAL

🎯 TIER A

🎯 TIER B

🎯 TIER C

✓ ANSWER SHEET — TEACHER COPY

STUDENT NAME

Any student name

DATE

Jul 6, 2026

TIER

A / B / C

📖 PART 1 OF 4 · REVIEW

🕒 5 MIN

✓ ANSWER KEY

Quick Loop Check — Before You Build

✓ ANSWER KEY — QUICK LOOP CHECK

1. What does this code do? (fill in the blanks)

This draws a square with 4 sides.

2. Which loop runs forever until you press Stop?

→ **forever** (the gold Control block)

3. Name one real-life loop from today:

e.g. brushing teeth · blinking · breathing · chewing

4. Which Scratch block color has loops?

→ **Gold / Yellow-Orange (Control category)**

🔧 PART 2 OF 4 · BUILD IN SCRATCH

🕒 15 MIN

✓ ANSWER KEY

Choose Your Tier Path — Model Answers

🎯 TIER A — EXPLORER

Make your sprite draw a **repeating shape** or **animation** using a loop.

Key formula: $\text{sides} \times \text{turn}^\circ = 360^\circ$

- **repeat 4** → **square** ($4 \times 90^\circ = 360^\circ$)
- Change to **repeat 3** → **triangle** (120°)
- **repeat 6** → **hexagon** (60°)

- ☒ Opened my Scratch project
- ☒ Sprite draws a repeating shape
- ☒ Tried changing the repeat number

🎯 TIER B — BUILDER

Build an animation using **both repeat** and **forever** in the same project.

- **forever** → **move + bounce + wait**
- **repeat 360** → **turn 1° + wait 0.01**
- Code savings: **2 blocks $\times$ 360 = saved 358 lines!**

forever = unknown duration · repeat N = known count

- ☒ Used both repeat & forever
- ☒ Something visibly repeats
- ☒ Counted code savings

🎯 TIER C — INNOVATOR

Build a **nested loop** generative art piece or a **loop + variable** animation.

- **repeat 12** → outer (12 petals)
- **repeat 4** + turn 30° → **each petal rotated**
- **Random pen color each outer loop**
- **Total: $12 \times 4 = 48$ inner iterations**

- ☒ Nested loops working
- ☒ Project looks unique/complex
- ☒ Saved project link (below)

📖 PART 3 OF 4 · PROJECT LOG

🕒 5 MIN

✓ ANSWER KEY

Record What You Built

📝 FILL IN AFTER BUILDING

What did your project do? **Drew a repeating shape / bouncing animation**

Loop used? **repeat N / forever / both**

Number in repeat? **4 (sq), 3 (tri), 6 (hex), 360 (spin), 12 (mandala)**

Blocks inside loop? **move 100, turn 90 (square)**

Changed a number? **e.g. reduced move 100→60 to fit screen**

Code Savings 📄

My loop has 2 blocks. Without a loop: $2 \times 4 = 8$ lines. I saved 6 lines! 🎉

📐 SKETCH YOUR PROJECT'S OUTPUT

[Square / hexagon / spiral / mandala sketch area]

Describe it in one sentence:

My Scratch project uses a loop to draw a square by repeating move 100 + turn 90 four times.

Something that didn't work & how I fixed it:

Sprite went off-screen → reduced move value from 100 to 60.

Scratch link (optional, Tier C encouraged):

scratch.mit.edu/projects/...

Connect Your Loop to Real AI

🧠 HOW AI REALLY TRAINS — ANNOTATED

```
# Inside every AI (simplified):  
repeat 10000 # ← training loops (epochs) ← THIS IS A LOOP!  
  look_at(example) # reads one training example  
  adjust(mistake) # updates weights = LEARNING  
# After 10,000 loops → smarter AI!
```

🔥 Each pass through all training data = one epoch . Real models may train for 50+ epochs on millions of examples!

1. How is your Scratch loop like AI training?

My Scratch repeat 4 loop runs move+turn 4 times to build a shape.
AI's training loop runs look_at+adjust 10,000 times to build intelligence. Both use repetition — without the loop, neither the shape nor the AI would work.

2. If AI trained only 3 loops instead of 10,000?

The AI would barely learn — it would memorize those 3 examples but fail on new data ("underfitting"). Like drawing only 1 side of a square: you need all 4 repeats or the shape never closes.

📊 GRADING RUBRIC

CATEGORY	FULL CREDIT	PARTIAL CREDIT	POINTS
🔧 Build	Loop used; something repeats visibly; saved in Scratch	Loop attempted but may not run correctly	4 pts
🗉 Log	All table rows filled with specific details	Most rows filled; some answers vague	3 pts
📊 Savings	Correctly counted lines saved by the loop	Attempted; minor calculation error	1 pt
📝 AI	2+ reflections in full sentences with real AI connection	1 reflection; connection to AI is vague	2 pts

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Summer 2026 AI Coding Program · Modern Chinese School · Session 4 Homework · scratch.mit.edu · Bring questions Monday!